

2021 JC2 PRELIMINARY EXAMINATION

BIOLOGY

9744/01

PAPER 1

Multiple Choice

21 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and identification number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

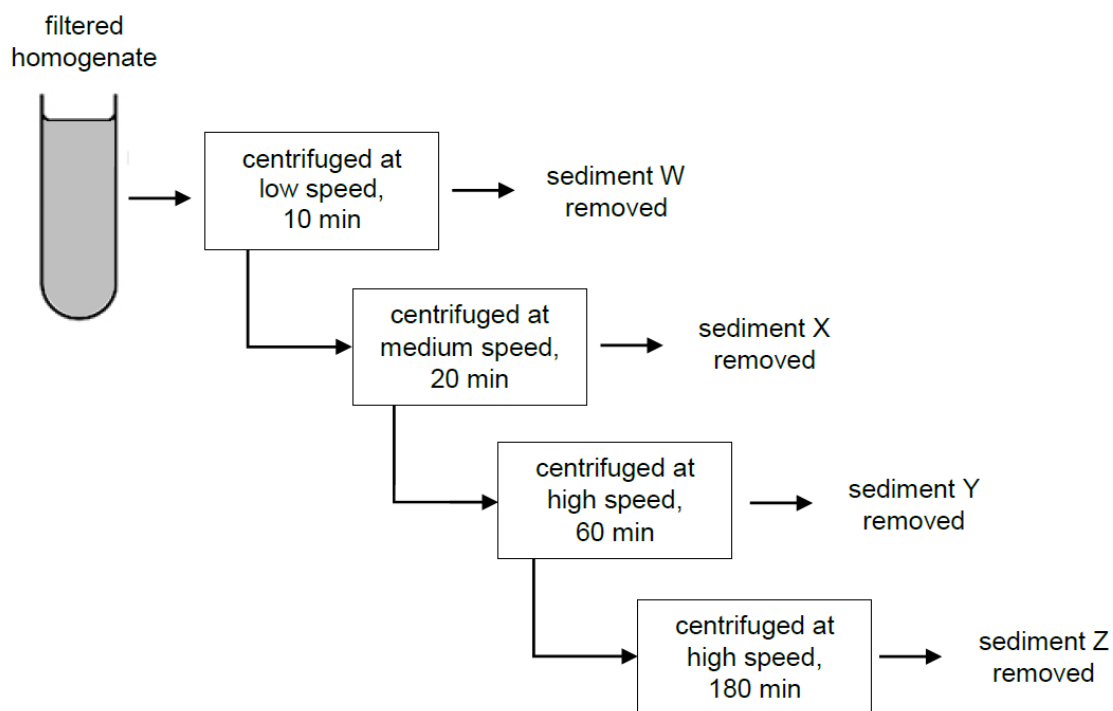
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **20** printed pages.

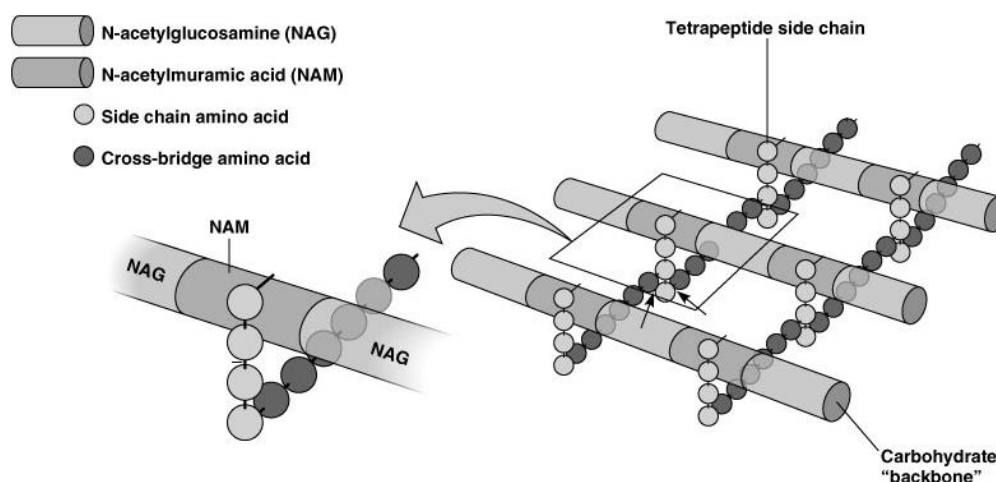
- 1 The diagram shows how a piece of homogenised mammalian tissue is subjected to cell fractionation.



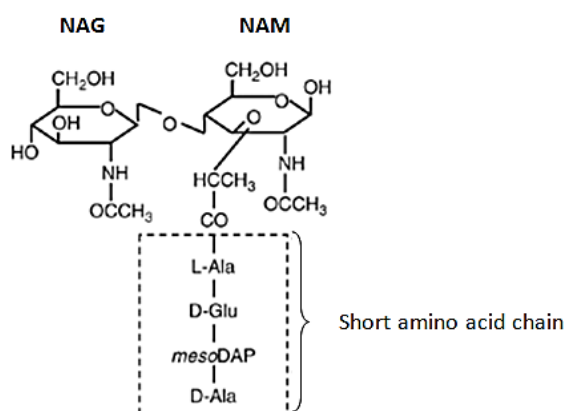
Which of the following statements is correct?

- A Sediment W contains the smallest organelle since it was centrifuged at the lowest speed.
- B Sediment X contains an organelle that is involved in the production of ATP.
- C Sediment Y contains fragments of plasma membrane which have had their phosphoester bonds broken by homogenisation.
- D Sediment Z contains double-stranded circular DNA released when organelles from sediment W were lysed.

- 2 The peptidoglycan layer in bacterial cell walls is a crystal lattice structure formed from linear chains of two alternating amino sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM). NAG and NAM differ from glucose only at the C2 and C3 positions.



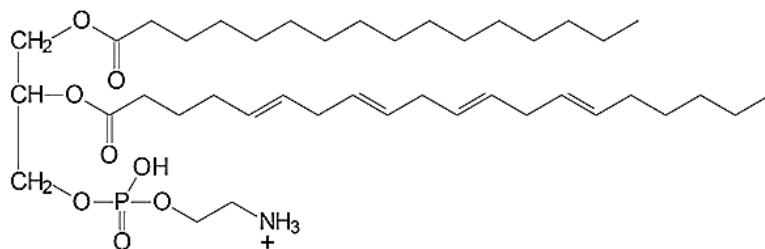
Each NAM is attached to a short amino acid chain. The figure below shows the molecular structure of NAM and NAG.



Which of the following statements correctly compares peptidoglycan and cellulose?

	similarity	difference
A	both contain β (1,4) glycosidic bonds	two different types of monomers present in cellulose; one type of monomer present in peptidoglycan
B	monomers are rotated 180° to each other	cross-links between cellulose chains are hydrogen bonds; cross-links between carbohydrate chains in peptidoglycan are peptide bonds
C	bundling of chains increases tensile strength	one type of bond found in cellulose fibrils; two types of bonds found in peptidoglycan
D	both comprise chains lying parallel to one another	β -glucose is found in cellulose; α -glucose is found in peptidoglycan

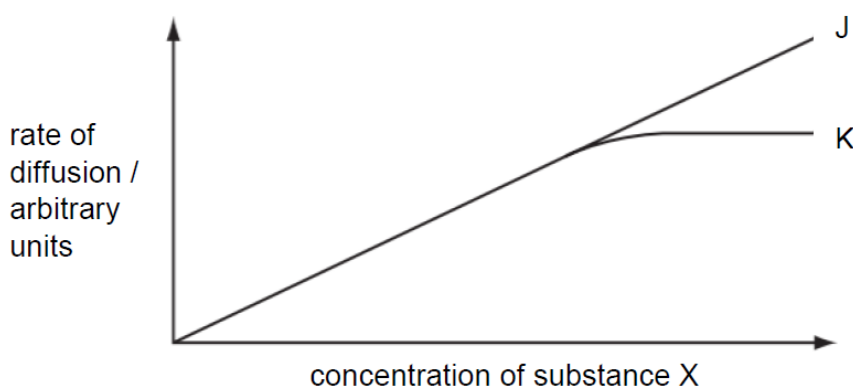
- 3 The structure of phosphatidylcholine, a common membrane phospholipid, is shown.



Which combination correctly describes the synthesis and property of one molecule of phosphatidylcholine?

	number of water molecules eliminated during synthesis	property
A	3	amphipathic
B	2	amphipathic
C	3	amphoteric
D	2	amphoteric

- 4 The graph shows rates of simple diffusion and facilitated diffusion of substance X across the cell surface membrane.



Which row is correct?

	graph K	explanation
A	facilitated diffusion	limited by the number of protein channels in the cell surface membrane
B	simple diffusion	allows the concentrations of substance X inside and outside the cell to be equal in a shorter time
C	facilitated diffusion	allows dynamic equilibrium to be achieved in a shorter time
D	simple diffusion	limited by the number of transient gaps in the cell surface membrane

5 The following statements are about the loading of haemoglobin with oxygen.

- 1 As the concentration of oxygen increases, the loading of oxygen increases.
- 2 Haemoglobin that has not bound to oxygen has a higher affinity for 2,3-bisphosphoglycerate (2,3-BPG) than for oxygen. 2,3-BPG is present in all red blood cells.
- 3 Binding of some oxygen to haemoglobin increases the affinity of haemoglobin for additional oxygen.
- 4 Binding of oxygen to haemoglobin causes any bound carbon dioxide or bound 2,3-BPG to be expelled.
- 5 Low pH reduces the affinity of haemoglobin for oxygen.

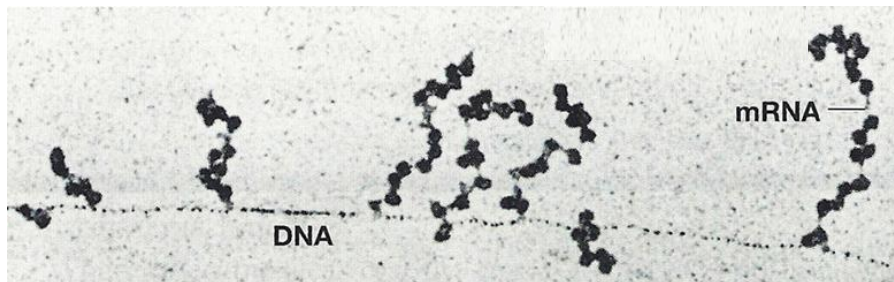
What explains the effect of a low concentration of oxygen on the loading of oxygen by haemoglobin?

- A** The affinity of haemoglobin for oxygen decreases at low concentrations of oxygen since carbon dioxide replaces oxygen, causing the pH to fall.
- B** Carbon dioxide remains bound to amino acids in the haemoglobin molecule and therefore makes these sites unavailable for the binding of oxygen.
- C** At low oxygen concentrations, oxygen is unable to compete with 2,3-BPG for binding sites on haemoglobin.
- D** At low oxygen concentrations, carbon dioxide expels 2,3-BPG from haemoglobin, decreasing the affinity of haemoglobin.

6 Which statement regarding the enzymatic activity of catalase under the following conditions is correct?

- 1 Addition of a non-competitive inhibitor
 - 2 Addition of pH 13 buffer solution
 - 3 Incubation at 10°C
- A** The hydrogen and ionic bonds between R groups of residues are broken in conditions 2 and 3, hence the rate of reaction decreases.
 - B** The catalytic and binding residues in all active sites are affected in condition 1, hence the rate of reaction decreases.
 - C** The chances of effective collisions to form enzyme-substrate complexes are low in condition 3, hence the rate of reaction decreases.
 - D** The changes in the 3D conformation of enzymes are irreversible in conditions 1, 2 and 3, hence the rate of reaction decreases.

- 7 Which of the descriptions of RNA is incorrect?
- A Ribosomal RNAs catalyse the formation of phosphodiester bonds.
 - B Transfer RNAs bind with messenger RNA during translation.
 - C Small nuclear RNAs bind with proteins to form spliceosomes.
 - D Messenger and transfer RNAs can contain secondary structures.
- 8 Protein synthesis in a prokaryote is shown in the micrograph below. Several mRNAs are evident.



How many of the following can be determined in the micrograph above?

- 1 the locations of ribosomes
- 2 the locations of RNA polymerases
- 3 the locations of the 5' ends of the mRNAs
- 4 the locations of polypeptides undergoing synthesis

A 1 **B** 2 **C** 3 **D** 4

- 9 The induction from the lysogenic to the lytic cycle of a particular temperate phage can be determined by observing the gene expression of β -galactosidase in *Escherichia coli*. This induction is known to occur over a temperature range of 32°C to 52°C.

β -galactosidase is made up of two peptides, lac Z- α and lac Z- Ω , neither of which is active by itself. However when both are present together, they assemble into a functional enzyme. Mutant *E. coli* containing a deletion in the *lac* operon does not produce lac Z- α .

In an experiment, phages with a *lac Z- α* gene under the control of a viral promoter are artificially synthesised. Upon transduction with these phages, mutant *E. coli* are able to produce functional β -galactosidase.

The amount of β -galactosidase produced in these recipient bacterial cells can then be measured using a colourless chemical known as X-gal. Hydrolysis of X-gal by β -galactosidase will result in the formation of a blue product. The colour intensity of this product can then be determined using a colorimeter.

Which of the following statements regarding the experiment are correct?

- 1 The higher the temperature, the higher the absorbance units recorded.
- 2 Expression of lac Z- Ω can occur during the lysogenic cycle.
- 3 Expression of lac Z- α can occur during the lytic cycle.
- 4 Lactose is required for the expression of lac Z- α .

A 1, 2 and 3

B 1 and 4

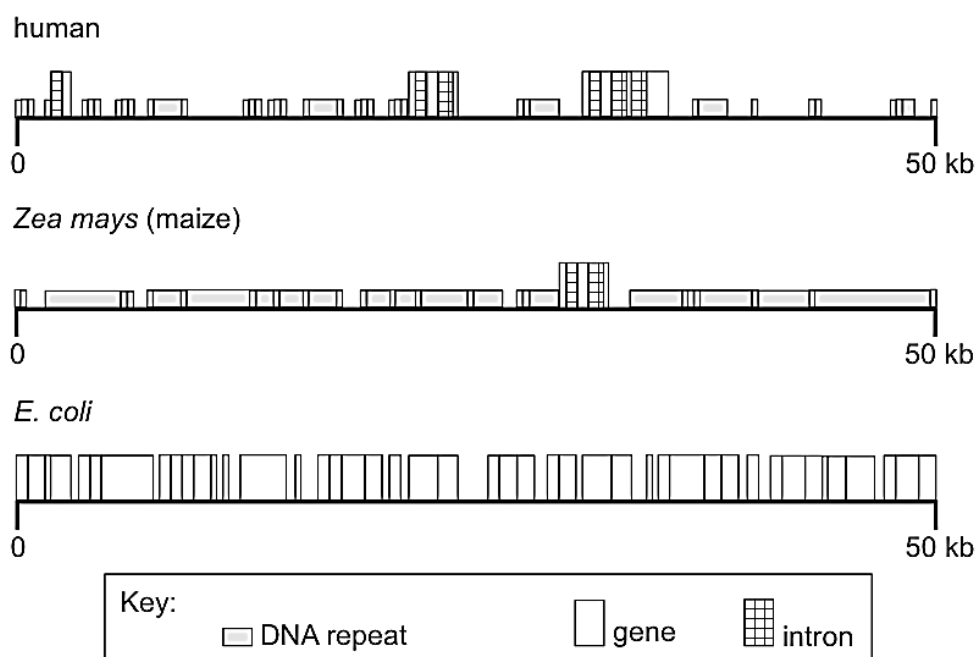
C 2 and 3

D 3 and 4

10 Which set of statements correctly describes the respective processes observed in bacteria?

	binary fission	transduction	conjugation
A	spindle fibres needed to separate chromosomes	may involve bacterial genes near to the integrated prophage	involves transfer of single-stranded DNA to recipient cell
B	spindle fibres needed to separate chromosomes	may involve a small piece of degraded host DNA	involves transfer of double-stranded DNA to recipient cell
C	daughter cells inherit bacterial chromosome and plasmids	may involve a small piece of degraded host DNA	involves transfer of single-stranded DNA to recipient cell
D	daughter cells inherit bacterial chromosome and plasmids	may involve bacterial genes near to the integrated prophage	involves transfer of double-stranded DNA to recipient cell

11 The diagram shows 50 kb segments of the human, *Zea mays* and *E. coli* genomes.



Which of the following statements can be concluded from the above results?

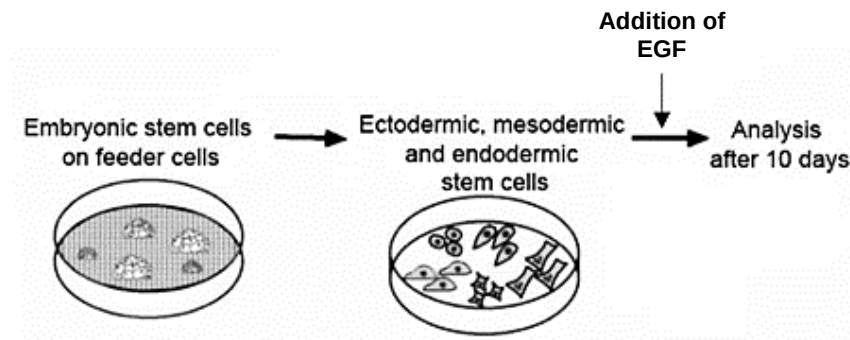
- A** More complex organisms have lower gene density.
- B** Organisms with smaller chromosome numbers have higher gene density.
- C** *Zea mays* has a higher density of DNA repeats as compared to human and *E. coli*.
- D** The presence of introns in the DNA of eukaryotes allows alternative splicing to occur to synthesise as many proteins as prokaryotes.

12 The regulation of gene expression involves control elements.

Assuming that only control elements X, Y and Z are involved, which of the following shows a probable relationship for tissue-specific gene expression within pancreas cells (P) and immature red blood cells (R) in eukaryotes?

	contains gene for glucagon	expresses genes for haemoglobin	regulated by control element X	regulated by control element Y	regulated by control element Z
A	P and R	P and R	P and R	P only	P only
B	P and R	R only	P and R	R only	R only
C	P only	R only	P only	R only	R only
D	P only	R only	P and R	P and R	P and R

- 13** Embryonic stem cells are widely used in treating a variety of acquired or hereditary diseases. An experiment investigated the effect of the addition of an epidermal growth factor (EGF) on stem cell differentiation in ectodermic, mesodermic and endodermic embryonic stem cells.



The table shows the expression of specific genes after the addition of EGF in the embryonic stem cells that code for:

- keratin, an important protein that forms a protective layer on the outside of the skin
- α -globin, a component of haemoglobin that aids red blood cells in carrying oxygen throughout the body
- insulin, a hormone produced by the pancreas that controls blood glucose levels by increasing glucose uptake into cells

type of embryonic stem cell	expression of specific gene after ten days		
	keratin	α -globin	insulin
ectodermic	+	–	–
mesodermic	–	+	–
endodermic	–	–	–

+ : gene is expressed

– : gene is not expressed

Which statement(s) is/are valid based on the information given?

- 1 The epidermal growth factor receptor is present only on ectodermic stem cells.
- 2 The addition of the growth factor initiated signal transduction, leading to the transcription of various specific genes and translation of these mRNAs.
- 3 The insulin gene is not expressed in the three types of embryonic stem cells as it is transcriptionally active in B cells only.

A 1, 2 and 3

B 2 and 3 only

C 2 only

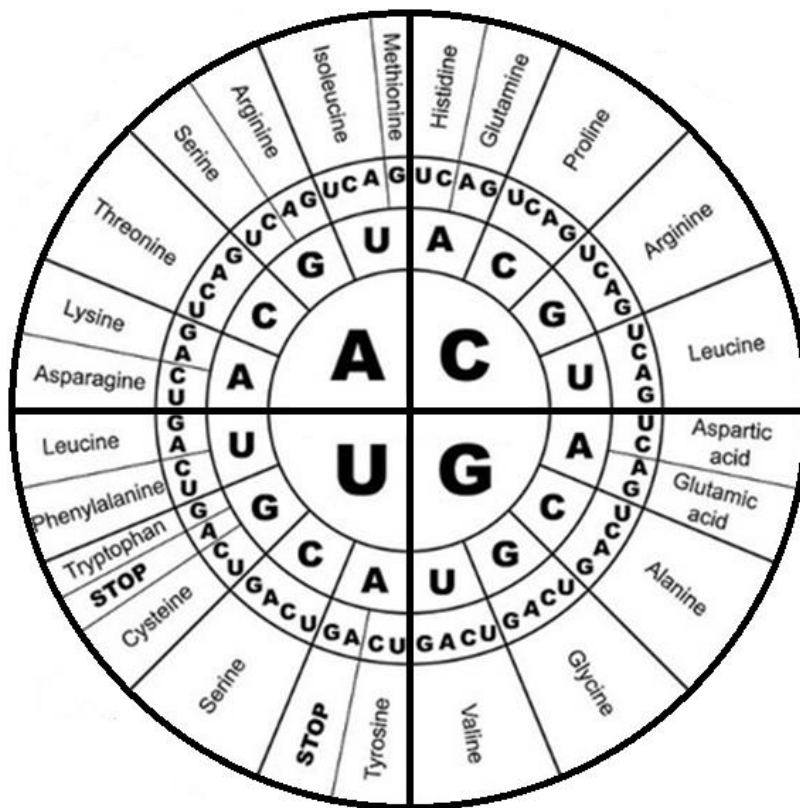
D 3 only

- 14 When a polymerase chain reaction amplification was performed on a sample of human genomic DNA, multiple products of varying sizes were obtained, including one of the expected size.

Which of the following modifications to the procedure is the most likely to eliminate the extra PCR products?

- A raising the annealing temperature from 52°C to 58°C
- B raising the elongation temperature from 70°C to 76°C
- C decreasing the number of cycles from 30 to 20
- D decreasing the amount of template DNA from 1.0 µg to 0.5 µg

- 15 The mRNA codons for amino acids are shown below.



A mutagen causes the adenine in DNA to pair with cytosine during transcription.

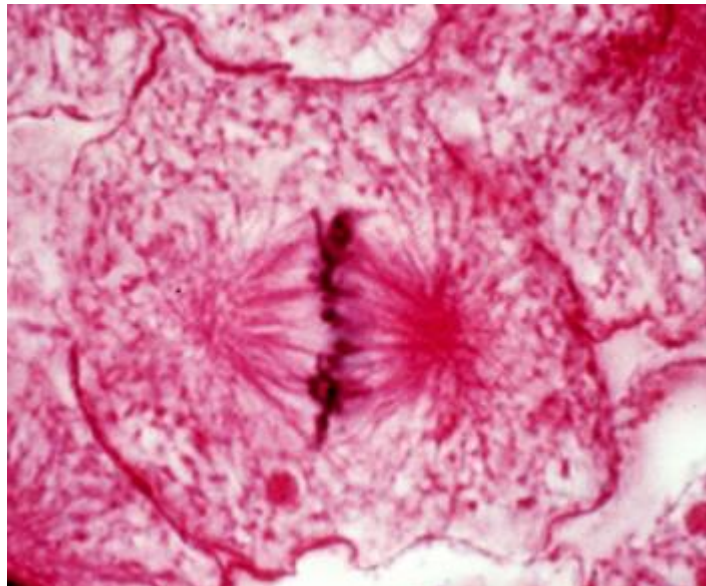
Which tripeptide will be synthesised when the DNA fragment ACCGTCAAT is used in protein synthesis in the presence of this mutagen?

- A proline-valine-proline
- B arginine-glutamine-proline
- C tryptophan-glutamine-leucine
- D threonine-valine-asparagine

- 16** A small proportion of men have the genotype XYY. Such genotypes are usually the consequence of non-disjunction occurring during meiosis. Non-disjunction results from a failure of chromosomes to separate correctly.

In which gamete and at which stage of meiosis must this non-disjunction occur?

- A** an egg produced by non-disjunction in meiosis I
 - B** an egg produced by non-disjunction in meiosis II
 - C** a sperm produced by non-disjunction in meiosis I
 - D** a sperm produced by non-disjunction in meiosis II
- 17** The micrograph below shows a stage of cell division immediately after fertilisation to form a zygote of an Atlantic freshwater fish.



Which of the following sets of terms correctly describes the stage shown above?

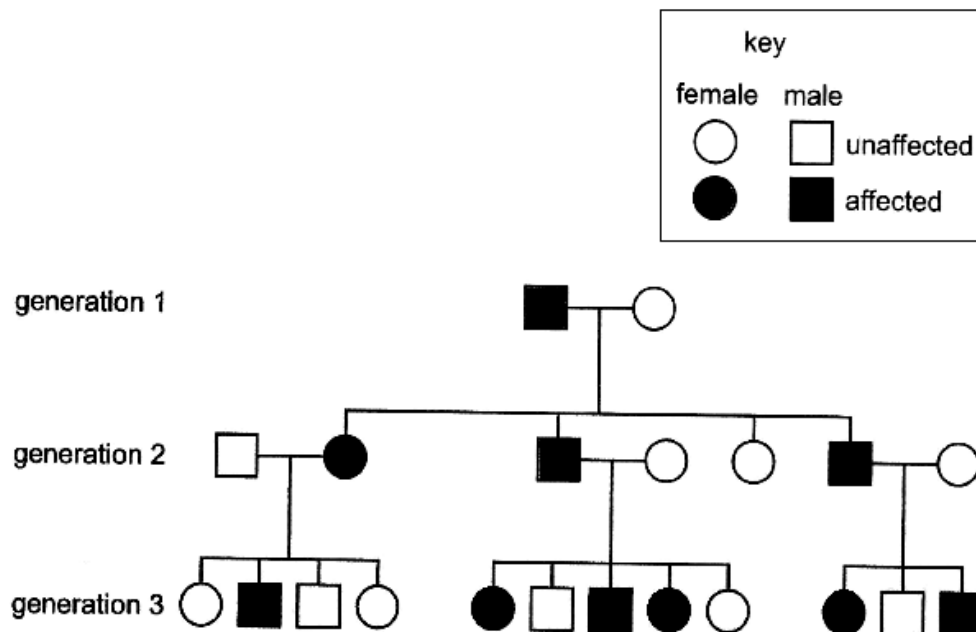
- A** homologous chromosomes are present, chromosomes are fully condensed
- B** chromosome number has doubled, cellular poles are established
- C** kinetochore microtubules have attached, homologous chromosomes have paired up
- D** cleavage furrow has formed, chromosomes are found at a single location

- 18** Melanoma is a form of skin cancer that begins in cells called melanocytes that control pigment production in the skin.

Which of the following correctly compares a cancerous and a normal melanocyte?

	similarity	difference
A	both cells arise from pre-existing cells	the cell cycle of cancerous melanocytes is arrested at either the S phase or M phase
B	some genes are switched off in both cells	cancerous melanocytes continue to divide even when they are tightly packed
C	both cells contain well-defined nuclei	cancerous melanocytes are always in the M phase of the cell cycle
D	active telomerase is present in both cells	more gene mutations are evident in cancerous melanocytes

- 19** The pedigree shows the inheritance of a dominant genetic condition in a family for three generations.



Which evidence indicates that this genetic condition is autosomal?

- A** Affected females always have affected sons.
- B** Affected males do not pass it on to their sons.
- C** Affected parents always have affected offspring.
- D** Males and females are equally affected.

20 The phenotype of the coat of a small mammal is controlled by two genes.

- The gene for hair colour has two alleles, **B** giving brown hair and **b** giving cinnamon hair.
- The gene for hair shape has two alleles, **C** giving curly hair and **c** giving straight hair.

A number of cinnamon, straight-haired females were crossed with brown, curly-haired males who were homozygous for both genes. All the F1 offspring showed the brown, curly-haired phenotype.

The F1 offspring were back-crossed to animals with their mother's genotype. This resulted in 1000 offspring of four different genotypes.

The table shows some of the results.

genotype	number
BbCc	R
Bbcc	S
P	160
Q	340

Which row correctly identifies the missing genotypes (P and Q) and expected offspring numbers (R and S)?

	P	Q	R	S
A	bbCc	bbcc	160	340
B	bbCc	bbcc	340	160
C	bbcc	bbCc	160	340
D	bbcc	bbCc	340	160

- 21** In garden peas, the allele **T** codes for the terminal position of flowers and is dominant over the allele **t**, which codes for the axil position.

In an experiment, two garden pea plants, both heterozygous for terminal flowers, produced 77 offspring with flowers at the terminal position and 43 offspring with flowers at the axil position.

A χ^2 test was done to determine whether the results of the experiment follow the expected pattern of inheritance.

Given that the χ^2 value at 1 degree of freedom and a probability of 0.05 is 3.84, which of the following conclusions can be drawn from the experiment and the χ^2 test?

Note:
$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where O is the observed value, E is the expected value and Σ is the 'sum of'

- A** Since the χ^2 value obtained from the experiment is greater than 3.84, flower position in garden peas does not involve a single pair of segregating alleles at gene locus T.
- B** Since the χ^2 value obtained from the experiment is greater than 3.84, flower position in garden peas involves a single pair of segregating alleles at gene locus T.
- C** Since the χ^2 value obtained from the experiment is less than 3.84, flower position in garden peas does not involve a single pair of segregating alleles at gene locus T.
- D** Since the χ^2 value obtained from the experiment is less than 3.84, flower position in garden peas involves a single pair of segregating alleles at gene locus T.
- 22** When the source of carbon dioxide is removed from a photosynthesising leaf, this can potentially affect the concentration of certain substances in the leaf.

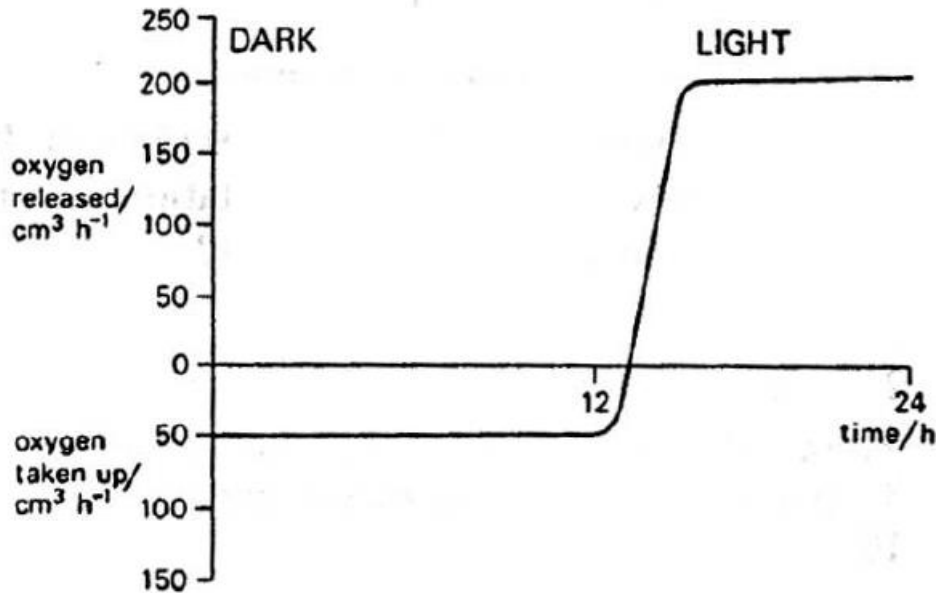
Which of the following represents how the concentration of the substances below could change?

	ATP	ribulose biphosphate	phosphoglyceric acid (PGA)
A	no change	decreases	no change
B	no change	decreases	decreases
C	decreases	increases	no change
D	increases	increases	decreases

- 23** Data was obtained relating to the rates of oxygen release and uptake in plants.

The plants were placed in the dark for 12 hours followed by 12 hours in the light. Temperature was kept constant throughout the experiment.

The results are shown in the graph below.



Which of the following is the most accurate estimate of the total volume of oxygen used by the plant for respiration during the 24 hours of the experiment?

- A** 600 cm³ **B** 1200 cm³ **C** 1800 cm³ **D** 3000 cm³

- 24** The following compounds occur in cellular respiration.

- 1 glyceraldehyde 3-phosphate
- 2 pyruvate
- 3 glucose
- 4 acetyl coA
- 5 oxaloacetate
- 6 carbon dioxide

Which shows the correct order in which these compounds first occur during cellular respiration?

- A** 3, 1, 2, 6, 4, 5
B 3, 1, 2, 4, 5, 6
C 3, 2, 4, 6, 5, 1
D 3, 2, 4, 5, 1, 6

- 25** Your teacher has never had much energy. He goes to a doctor for help and is sent to the hospital.

After some laboratory tests, it was discovered that his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal.

Which of the following is the best explanation for his condition?

- A** His diet is excessively high in lipids and proteins, and low in carbohydrates.
- B** His cells contain an inhibitor that prevents his mitochondria from using oxygen.
- C** The transport proteins on the envelopes of his mitochondria are defective.
- D** Reduced NAD produced during glycolysis cannot enter his mitochondria.

- 26** Four proteins isolated from a human cell were investigated for their involvement in cell signalling pathways.

	protein W	protein X	protein Y	protein Z
transmembrane domain	-	+	+	-
DNA binding domain	-	-	-	+
enzymatic domain	+	-	+	-

+ : domain is present

- : domain is absent

Which of the following correctly identifies the four proteins?

	protein W	protein X	protein Y	protein Z
A	G-protein linked receptor	G protein	tyrosine kinase receptor	activator
B	G protein	tyrosine kinase receptor	G-protein linked receptor	activator
C	activator	G-protein linked receptor	tyrosine kinase receptor	G protein
D	G protein	G-protein linked receptor	tyrosine kinase receptor	activator

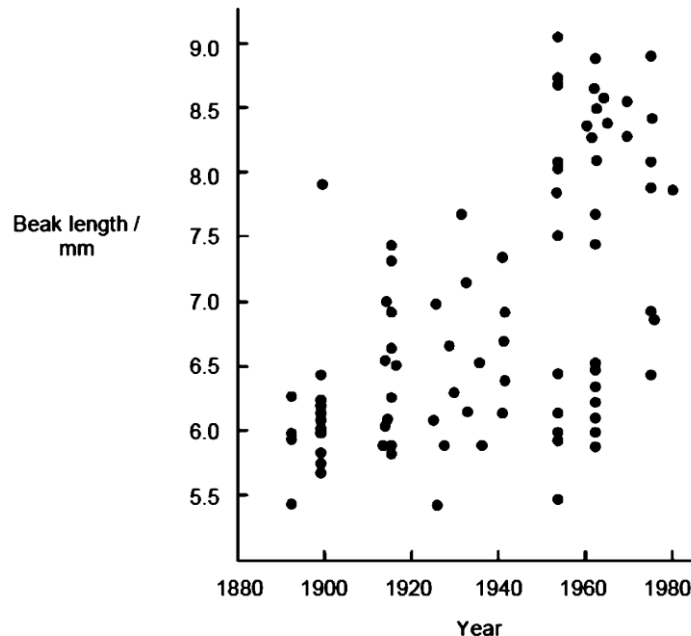
- 27 Recent DNA studies have examined the skeletal remains of Europeans buried during the plagues of the Roman Empire, the Middle Ages, the seventeenth and eighteenth centuries and in modern times. These plague outbreaks varied in their symptoms and severity. Despite these differences, the studies suggest that these plagues were all caused by the bacterium *Yersinia pestis*.

Which statement does **not** describe a feature that could contribute to the evolution of *Y. pestis* through natural selection?

- A Bacteria from the various strains of *Y. pestis* have different genotypes which could account for the changes in the symptoms and severity of the disease over the centuries.
- B Bacteria within each strain of *Y. pestis* have the same DNA sequence but, depending on their interaction with the human host, can cause different symptoms with a variety of consequences.
- C Changes in the genome of *Y. pestis* over the centuries may be associated with changes in its environment, including the changing genetic characteristics of human hosts.
- D Two DNA sequences that significantly increase the severity of the disease have been found in plasmids that replicate independently of the rest of the bacterial DNA.

- 28** The soapberry bug, *Jadera haematoloma*, uses its long beak to penetrate the fleshy fruit of the native soapberry tree to feed on the seeds at the centre. The bug also feeds on the fruit of the golden rain tree, which is a non-native species introduced to the area.

Investigators measured the beak length of the soapberry bugs over eighty years. The results are shown in the graph below.



In neighbouring regions, the fruits of introduced plants have also been fed on by soapberry bugs. Male and female soapberry bugs from different regions can interbreed. Evidence indicates that genetic isolation between some of these populations is gradually occurring.

The situation that would lead to an increase in genetic isolation would be if

- A** different types of host plants have fruiting seasons which do not overlap.
- B** chemicals released by the female bugs attract male bugs from neighbouring populations.
- C** the soapberry tree is commonly found where the bugs are distributed and each tree produces large amounts of fruit.
- D** male bugs new to a region are reproductively active, whereas female bugs need to feed before becoming reproductively active.

- 29** Which statement(s) describe(s) both B lymphocytes and T lymphocytes?

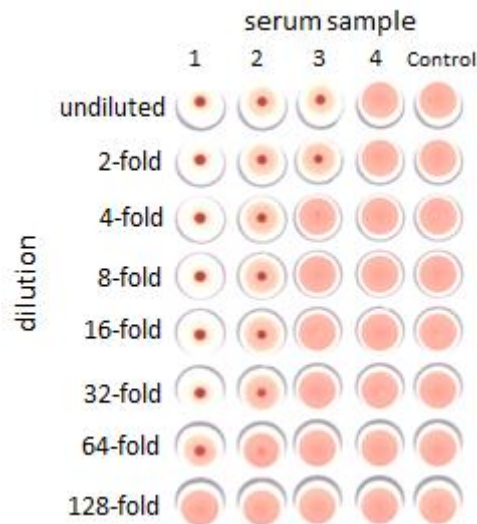
- 1 They contain specific protein receptors in their cell surface membranes.
- 2 They differentiate into antibody-secreting cells.
- 3 They divide by mitosis.
- 4 They secrete cytokines.

- A** 1, 3 and 4 **B** 2 and 3 **C** 3 and 4 **D** 1 only

- 30** The effectiveness of a vaccine is measured by the concentration of antibodies produced that can bind to specific antigens in the vaccine.

Blood from an individual is taken at different times after vaccination and the blood serum containing antibodies is extracted. The serum is then subjected to a serial dilution where the concentration is halved with each step of the dilution. Each diluted blood serum is added to a well containing a fixed concentration of antigens. At a sufficiently high antibody concentration, a visible dark spot is formed due to a high level of antigen-antibody agglutination.

The diagram shows the results of an investigation on the effectiveness of the first dose of the vaccine and the second dose given after six months. Four serum samples were taken at different time points.



Which serum samples correspond to the different time points after the first dose and the second dose were given?

	two days after first dose is given	immediately before second dose is given	two days after second dose is given	six months after second dose is given
A	1	2	4	3
B	4	2	3	1
C	4	3	1	2
D	1	4	2	3

Paper 1 Answer Scheme

Qns	Ans
1	B
2	B
3	A
4	A
5	C
6	C
7	A
8	D
9	C
10	C
11	C
12	B
13	C
14	A
15	B
16	D
17	A
18	B
19	D
20	B
21	A
22	D
23	B
24	A
25	C
26	D
27	B
28	A
29	A
30	C